

TEMPORAL AGGREGATION IN THE ARIMA PROCESS

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Abstract. The effect of temporal aggregation on ARIMA models is investigated. The paper discusses the change of model form resulting from aggregation. For the IMA model it is noted that reduction of model order may occur, due to aggregation, which takes an arbitrarily high order IMA (d, q) process to an IMA $(d, 0)$ process for the aggregates. For the AR process, we derive the exact order for the aggregate model and show that aggregation of an AR (p) series does not necessarily produce an ARMA (p, q) aggregate series as has been suggested in the literature. In particular, the AR order of AR (p) and ARIMA (p, d, q) can be reduced upon aggregation. The paper gives the general conditions under which this reduction of order may occur.

Keywords. Aggregation; aggregate model; ARIMA process; change of model form.

1. INTRODUCTION

In any scientific investigation, after making a conjecture or a proposition about the underlying model of a phenomenon, a scientist must test his proposition against what is observable. Suppose that based on some theoretical considerations, a scientist proposes the underlying model for a time series process in terms of a basic time unit t . In some fields, a scientist can test his proposition by designing an experiment and analyzing the data obtained from the experiment in terms of the same time unit t used in his proposition. However, in some cases controlled experiments may not be possible. Although time series observations are made available at some given equal time interval, often the interval cannot be chosen by the investigator. Thus, the assumed basic time unit may not be the same as the time unit for which data are observed. In many studies, data are often available only in aggregated form.

This paper discusses the effect of temporal aggregation on the model form of an ARIMA process. It is assumed in this paper that a given univariate time series, $x_t, t = -\infty, \dots, 0, \dots, \infty$, has an ARIMA (p, d, q) representation of a known form

$$\phi_p(B)(1-B)^d x_t = \theta_q(B) a_t \quad (1.1)$$

Here $\phi_p(B) = (1 - \phi_1 B - \dots - \phi_p B^p)$ and $\theta_q(B) = (1 - \theta_1 B - \dots - \theta_q B^q)$, are polynomials in B , the backshift operator, defined such that $Bx_t = x_{t-1}$. The white noise series a_t is a sequence of *iid* random variables with mean 0, and finite variance. Additionally, both $\phi_p(B)$ and $\theta_q(B)$ are assumed to have their roots outside the unit circle and to have no roots in common. Temporal aggregation means the construction of the aggregate series

$$y_t = \sum_{i=m(t-1)+1}^{mt} x_i. \quad (1.2)$$

Here m is fixed and is termed the order of aggregation. Thus, for example, if m is 3, y_t may be the quarterly sum of a monthly series x_t . The subject of this paper is then the analysis of the model form for the y_t series, from the knowledge of the model for x_t .

Authors who have investigated the effects of aggregation on various types of ARIMA models include Telser (1976), Amemiya and Wu (1972), Tiao (1972), Kendall (1973), Brewer (1973), Wei (1978), and Harvey (1981, p. 44). Telser (1967), and Amemiya and Wu (1972), dealt with the purely autoregressive, or AR(p) models. The result which they reported is that a purely autoregressive model AR(p) is transformed upon aggregation to a mixed model of form ARMA(p, q^*), where the roots of the aggregate model's AR polynomial are equal to the m th powers of those of the basic series model. Amemiya and Wu also gave the MA order, q^* , of the mixed model for the aggregates to be equal to $[p+1-(p+1)/m]$, where $[X]$ denotes the integer part of X . Moreover, they proved that the MA polynomial for the aggregate series is invertible; that is, the roots of the MA polynomial lie outside the unit circle. Brewer (1973) and Harvey (1981) discussed the effect of aggregation of ARMA(p, q) mixed models. The result which Brewer and Harvey gave was that the ARMA(p, q) process would, after aggregation, be transformed to an ARMA(p, q^*) process with q^* equal to $[p+1+(q-p-1)/m]$.

The AR models which Amemiya and Wu, and Telser dealt with, and the ARMA models treated by Brewer and Harvey are all stationary. The first author to deal directly with the effect of aggregation on a nonstationary ARIMA model was Tiao (1972) who examined the effect of aggregation on the integrated moving average IMA(d, q) model. Wei (1978) extended both Brewer and Tiao's results to the general multiplicative seasonal model and showed that for a given basic series of a seasonal autoregressive integrated moving average process ARIMA(p, d, q) $\times$ (P, D, Q)_s, the m th order aggregate series follows an ARIMA(p, d, q^*) $\times$ (P, D, Q)_s process where $q^* = [p+d+1+(q-p-d-1)/m]$ and $S = s/m$.

It is clear that the result published so far indicates that the autoregressive order p in AR(p) or ARIMA(p, d, q) models is unchanged upon aggregation. They fail to point out that their derived orders for the autoregressive and moving average polynomials are only maximum. In this paper we derive the exact order for the aggregate model. Specifically, in section 2, we derive the relationship between the autocovariance functions of a given time series and its aggregates.

In section 3, we discuss the aggregation of the IMA (d, q) model and show that the reduction of the MA order may occur due to aggregation, which takes an arbitrary high order q in the IMA (d, q) process to an IMA $(d, 0)$ process. In section 4, we discuss the aggregation of the AR (p) and ARIMA (p, d, q) processes and show that although the aggregate model is, in general, a mixed ARMA model, the autoregressive order p may be reduced upon aggregation. A general condition under which the order p is unchanged by aggregation is given. In this case, when the order of aggregation equals m , the roots of the autoregressive polynomial of the aggregate model are the m th powers of the roots of the basic series autoregressive polynomial. Finally, in section 5, some concluding remarks are given.

2. THE RELATIONSHIP OF AUTOCOVARIANCES BETWEEN THE BASIC SERIES AND ITS AGGREGATE

Let $\{x_t\}$ be the equally spaced basic series whose d th difference $w_t = (1 - B)^d x_t$ follows a covariance stationary process with a mean 0 and the autocovariance function $\gamma_w(k) = \text{cov}(w_t, w_{t-k})$. Define $Y_t = (1 + B + \dots + B^{m-1})x_t$. We have the aggregate series $y_t = Y_{mt}$ and $(1 - B)^d y_t = (1 - B^m)^d Y_{mt}$. Thus, by letting $u_t = (1 - B)^d y_t$, we have

$$\begin{aligned} u_t &= (1 - B)^d y_t = (1 - B^m)^d Y_{mt} \\ &= (1 + B + \dots + B^{m-1})^d (1 - B)^d (1 + B + \dots + B^{m-1}) x_{mt} \\ &= (1 + B + \dots + B^{m-1})^{d+1} w_{mt}. \end{aligned} \quad (2.1)$$

Hence, the d th differenced aggregate series $u_t = (1 - B)^d y_t$ is covariance stationary because it is a finite moving average of a stationary process w_t .

Let $\gamma_u(k)$ be the autocovariance function for the d th differenced aggregate series $\{u_t\}$. We have the following theorem.

THEOREM 2.1. *Let $w_t = (1 - B)^d x_t$ be a covariance stationary process with its k th autocovariance $\gamma_w(k)$. Let $y_t = (\sum_{j=0}^{m-1} B^j) x_{mt}$. Then the k th autocovariance of $u_t = (1 - B)^d y_t$ equals*

$$\begin{aligned} \gamma_u(k) &= (1 + B + \dots + B^{m-1})^{d+1} (1 + F + \dots + F^{m-1})^{d+1} \gamma_w(mk) \\ &= (1 + B + \dots + B^{m-1})^{2(d+1)} \gamma_w(mk + (d+1)(m-1)) \end{aligned} \quad (2.2)$$

where B is the back shift operator such that $Bw_t = w_{t-1}$ and $B\gamma_w(i) = \gamma_w(i-1)$, and $F = B^{-1}$ is the forward shift operator such that $Fw_t = w_{t+1}$ and $F\gamma_w(i) = \gamma_w(i+1)$.

PROOF. From (2.1) we have $u_t = (1 - B)^d y_t = (1 + B + \dots + B^{m-1})^{d+1} w_{mt}$. Let c_i be the coefficients of B^i and F^i in the polynomials $(1 + B + \dots + B^{m-1})^{d+1}$ and

$$(1 + F + \dots + F^{m-1})^{d+1}.$$

$$\begin{aligned} \gamma_u(k) &= E(u_t u_{t-k}) = E\{1 + B + \dots + B^{m-1}\}^{d+1} w_{mt} \\ &\quad \cdot (1 + B + \dots + B^{m-1})^{d+1} w_{m(t-k)}\} \\ &= E(C_0 w_{mt} + C_1 w_{mt-1} + \dots + C_{(d+1)(m-1)} w_{mt-(d+1)(m-1)}) \\ &\quad \cdot (C_0 w_{mt-mk} + C_1 w_{mt-mk-1} + \dots + C_{(d+1)(m-1)} w_{mt-mk-(d+1)(m-1)})\} \\ &= C_0\{C_0 \gamma_w(mk) + C_1 \gamma_w(mk+1) + \dots \\ &\quad + C_{(d+1)(m-1)} \gamma_w(mk + (d+1)(m-1))\} \\ &\quad + C_1\{C_0 \gamma_w(mk-1) + C_1 \gamma_w(mk) + \dots \\ &\quad + C_{(d+1)(m-1)} \gamma_w(mk + (d+1)(m-1) - 1)\} + \dots \\ &\quad + C_{(d+1)(m-1)}\{C_0 \gamma_w(mk - (d+1)(m-1)) + C_1 \gamma_w(mk - (d+1)(m-1) + 1) \\ &\quad + \dots + C_{(d+1)(m-1)} \cdot \gamma_w(mk)\} \\ &= C_0\{(1 + F + \dots + F^{m-1})^{d+1} \gamma_w(mk)\} \\ &\quad + C_1\{(1 + F + \dots + F^{m-1})^{d+1} \gamma_w(mk-1)\} + \dots \\ &\quad + C_{(d+1)(m-1)}\{(1 + F + \dots + F^{m-1})^{d+1} \gamma_w(mk - (d+1)(m-1))\} \\ &= (1 + F + \dots + F^{m-1})^{d+1}\{C_0 \gamma_w(mk) + \dots \\ &\quad + C_{(d+1)(m-1)} \gamma_w(mk - (d+1)(m-1))\} \\ &= (1 + F + \dots + F^{m-1})^{d+1} (1 + B + \dots + B^{m-1})^{d+1} \gamma_w(mk) \\ &= F^{(d+1)(m-1)} (1 + B + \dots + B^{m-1})^{2d+2} \gamma_w(mk) \\ &= (1 + B + \dots + B^{m-1})^{2(d+1)} \gamma_w(mk + (d+1)(m-1)). \quad \square \end{aligned}$$

Theorem 2.1 defines each autocovariance $\gamma_u(k)$ as a linear transformation of the autocovariances $\gamma_w(j)$ for $j = mk - (d+1)(m-1)$ to $j = mk + (d+1)(m-1)$. The coefficient in the linear transformation are found by expanding the polynomial $(1 + B + \dots + B^{m-1})^{2(d+1)}$. This is a useful relationship between the autocovariances of the basic series $\{x_t\}$ and those of the aggregate series $\{y_t\}$.

3. TEMPORAL AGGREGATION ON THE IMA (d, q) MODEL

The model which we consider in this section is of form,

$$w_t = (1 - B)^d x_t = \theta_q(B) a_t \quad (3.1)$$

where a_t is a series of i.i.d. random errors and $\theta_q(B)$ is a polynomial in B , with its roots outside the unit circle.

Now in order to discuss the effect of aggregation in the IMA (d, q) model we first develop the following notation. Expression (2.2) defines the autocovariances $\gamma_u(k)$ of u_t as a linear transformation of the autocovariances $\gamma_w(k)$. We can write this transformation in matrix form as

$$\begin{bmatrix} \gamma_u(0) \\ \gamma_u(1) \\ \vdots \\ \gamma_u(k) \\ \vdots \end{bmatrix} = A \begin{bmatrix} \gamma_w(-(m-1)(d+1)) \\ \gamma_w(-(m-1)(d+1)+1) \\ \vdots \\ \gamma_w(0) \\ \vdots \\ \gamma_w(km+(m-1)(d+1)) \\ \vdots \end{bmatrix} \quad (3.2)$$

where the coefficient matrix A is equal to

$$\begin{bmatrix} C & 0 & \cdots & & \\ 0_m & C & 0 & \cdots & \\ \vdots & & & & \\ 0_{m(k-1)} & & \cdots & C & 0 & \cdots \\ \vdots & & & & & \end{bmatrix}$$

Here 0_l is a $1 \times l$ vector of zeros and C is a $1 \times (2(d+1)(m-1)+1)$ vector of c_i where c_i is the coefficient of B^i in the polynomial $(1+B+\cdots+B^{m-1})^{2(d+1)}$. Since $\gamma_w(-k) = \gamma_w(k)$ for all k , we can, in the above matrix, delete the first $(m-1)(d+1)$ columns corresponding to $\gamma_w(-(m-1)(d+1)), \dots, \gamma_w(-1)$, by adding them to the columns corresponding to $\gamma_w((m-1)(d+1)), \dots, \gamma_w(1)$, respectively. This gives

$$\begin{bmatrix} \gamma_u(0) \\ \gamma_u(1) \\ \vdots \\ \gamma_u(k) \\ \vdots \end{bmatrix} = A_m^d \begin{bmatrix} \gamma_w(0) \\ \gamma_w(1) \\ \vdots \\ \gamma_w(km+(m-1)(d+1)) \\ \vdots \end{bmatrix} \quad (3.3)$$

where A_m^d is the matrix formed by this deletion of the first $(m-1)(d+1)$ columns and adding them to the proper remaining columns of the matrix A .

For the IMA (d, q) representation the only nonzero autocovariances of w_t are $\gamma_w(0), \gamma_w(1), \dots, \gamma_w(q)$. From expression (2.2) we see that $\gamma_u(l)$ is a weighted sum of those $\gamma_w(k)$ such that $ml-(d+1)(m-1) \leq k \leq ml+(d+1)(m-1)$. Thus if $l > q^* = [d+1+(q-d-1)/m]$, the $\gamma_u(l)$ will be identically zero so that the only nonzero autocovariances of u_t , $\gamma_u(k)$, are at most $\gamma_u(0), \gamma_u(1), \dots, \gamma_u(q^*)$, for $q^* = [d+1+(q-d-1)/m]$. Thus, we have the following theorem.

THEOREM 3.1. *Let $\{x_t\}$ be the basic series following the IMA (d, q) model given in (3.1). Then the m th-order aggregate series $y_t = (1+B+\cdots+B^{m-1})x_{mt}$ follows an IMA (d, Q) process with $Q \leq q^* = [d+1+(q-d-1)/m]$.*

Theorem 3.1 implies that one need only consider in equation (3.3) the first q^*+1 rows and $q+1$ columns of the infinite coefficient matrix A_m^d when dealing with the aggregation of an IMA (d, q) model. We call the submatrix consisting of these rows and columns $A_m^d(q)$ and we have

$$\begin{bmatrix} \gamma_u(0) \\ \gamma_u(1) \\ \vdots \\ \gamma_u(q^*) \end{bmatrix} = A_m^d(q) \begin{bmatrix} \gamma_w(0) \\ \gamma_w(1) \\ \vdots \\ \gamma_w(q) \end{bmatrix}. \quad (3.4)$$

EXAMPLE 3.1. To clarify the construction of the matrix $A_m^d(q)$ we exhibit it for the case when $q=3$, $m=2$, and $d=2$. The coefficients of the polynomial $(1+B+\dots+B^{m-1})^{2(d+1)}$ are now equal to 1 6 15 20 15 6 1. We will have $q^*=3$ and expression (3.2) becomes

$$\begin{bmatrix} \gamma_u(0) \\ \gamma_u(1) \\ \gamma_u(2) \\ \gamma_u(3) \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 & 6 & 15 & 20 & 15 & 6 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 6 & 15 & 20 & 15 & 6 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 6 & 15 & 20 & 15 & 6 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 6 & 15 & 20 & 15 & 6 & 1 \\ \vdots & & & & & & \vdots & & & & & \vdots & \end{bmatrix} \begin{bmatrix} \gamma_w(-3) \\ \gamma_w(-2) \\ \gamma_w(-1) \\ \gamma_w(0) \\ \gamma_w(1) \\ \gamma_w(2) \\ \gamma_w(3) \\ \gamma_w(4) \\ \gamma_w(5) \\ \gamma_w(6) \\ \gamma_w(7) \\ \gamma_w(8) \\ \gamma_w(9) \\ \vdots \end{bmatrix}$$

Now since $\gamma_w(-3)=\gamma_w(3)$, $\gamma_w(-2)=\gamma_w(2)$ and $\gamma_w(-1)=\gamma_w(1)$, and since $\gamma_w(k)=0$, for $k>3$ and $q^*=3$, expression (3.4) becomes

$$\begin{bmatrix} \gamma_u(0) \\ \gamma_u(1) \\ \gamma_u(2) \\ \gamma_u(3) \end{bmatrix} = \begin{bmatrix} 20 & 30 & 12 & 2 \\ 6 & 16 & 20 & 15 \\ 0 & 1 & 6 & 15 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \gamma_w(0) \\ \gamma_w(1) \\ \gamma_w(2) \\ \gamma_w(3) \end{bmatrix}.$$

It should be recognized that the q^* derived in theorem 3.1 is only an upper bound. To see that, let us consider the case when $m=3$ and the model for x_t is an MA (2). In this case, $q^*=1$. Then from (3.4) we have that

$$A_m^d(q) = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

and

$$\begin{bmatrix} \gamma_y(0) \\ \gamma_y(1) \end{bmatrix} = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} \gamma_x(0) \\ \gamma_x(1) \\ \gamma_x(2) \end{bmatrix}.$$

Now $\gamma_x(1) = (-\theta_1 + \theta_1\theta_2)\sigma_a^2$ and $\gamma_x(2) = -\theta_2\sigma_a^2$ so the $\gamma_y(1) = (\theta_1\theta_2 - \theta_1 - 2\theta_2)\sigma_a^2$. Hence if $\theta_2 = \theta_1/(\theta_1 - 2)$ then the aggregates will be uncorrelated. In fact, we will prove later in theorem 3.2 that there is no upper limit on the order of MA (q) models for which the third-order aggregates are uncorrelated. To prove that theorem, we need the following results.

As previously mentioned, an IMA (d, q) model is said to be invertible if the roots of the MA polynomial lie outside the unit circle. We make the following definition.

DEFINITION 3.1. A $(q+1)$ -vector, $\rho_w = (\rho_w(0), \dots, \rho_w(q))'$, is termed admissible for the autocorrelations of a MA (q) model if there exists an invertible MA (q) series

$$w_t = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q) a_t,$$

having the components of ρ_w as its k lag autocorrelations, where $\rho_w(k) = \gamma_w(k)/\gamma_w(0)$, for $k = 0, 1, \dots, q$. The autocovariances, γ_w , will be termed admissible if ρ_w is admissible. The admissible region for the autocorrelations of a MA (q) series will consist of all admissible ρ_w .

Using the fact that for every invertible MA process we can find a strictly positive power spectrum $f(\omega)$, and vice versa, we prove the following Lemma.

LEMMA 3.1. *The admissible region for the autocorrelations of an invertible MA (q) model is an open set in R^q .*

PROOF. From the above we see that it is sufficient to show the following: The set A defined as

$$A = \left\{ (\rho_1, \dots, \rho_q) \in R^q : 1 + \sum_{i=1}^q 2\rho_i \cos(i\omega) > 0 \text{ for all } 0 \leq \omega \leq \pi \right\}$$

is open. Let $\rho_0 = (\rho_1^0, \rho_2^0, \dots, \rho_q^0)'$ be an element of A . We show that if ρ_0 is admissible then any $\rho = (\rho_1, \dots, \rho_q)'$ in the open neighborhood $\{\rho \in R^q : |\rho_i - \rho_i^0| < l/4q \text{ for all } i\}$ is also admissible when $l = \min_{0 \leq \omega \leq \pi} \{1 + \sum_{i=1}^q 2\rho_i^0 \cos(i\omega)\}$. We note that l is positive because a continuous function takes on its minimum over a closed bounded set. Let ρ be any member of the open neighborhood. Now since $\max_{\omega \in [0, \pi]} |\cos(i\omega)| \leq 1$ for any i we have that if $|\rho_i - \rho_i^0| < l/4q$ then $|2\rho_i \cos(i\omega) - 2\rho_i^0 \cos(i\omega)| < l/2q$ for all $0 \leq \omega \leq \pi$. And

$$\begin{aligned} l/2 &> \sum_{i=1}^q |2\rho_i \cos(i\omega) - 2\rho_i^0 \cos(i\omega)| > \left| \sum_{i=1}^q 2\rho_i \cos(i\omega) - 2\rho_i^0 \cos(i\omega) \right| \\ &= \left| 1 + \sum_{i=1}^q 2\rho_i \cos(i\omega) - 1 - \sum_{i=1}^q 2\rho_i^0 \cos(i\omega) \right| \quad \text{for all } 0 \leq \omega \leq \pi. \end{aligned}$$

Since l is the minimum value of $1 + \sum_{i=1}^q 2\rho_i \cos(i\omega)$ it follows that $1 + \sum_{i=1}^q 2\rho_i \cos(i\omega) > 0$ for all $0 \leq \omega \leq \pi$. Hence ρ is contained in A and A is open. $\square$

With the above result, we can prove the following theorem.

THEOREM 3.2. *Let $y_t = (1 + B + \cdots + B^{m-1})x_{mt}$ have an IMA (d, q^*) representation. If there exists an IMA (d, q) model for x_t with admissible autocorrelations, $\rho_w(k) = \gamma_w(k)/\gamma_w(0)$, which is transformed to the model for y_t , then there exists no upper bound on the MA order of an IMA model for x_t , which is transformed by aggregation to the model for y_t .*

PROOF. Suppose that the theorem is false and the q is the largest number for which an IMA (d, q) model for x_t can be found, which is transformed by aggregation to the model for the y_t . Clearly $q^* \leq [d+1+(q-d-1)/m]$. Let $\bar{q}$ be greater than $d+1+q[m/(m+1)]$ so that $\bar{q}-q > [d+1+(\bar{q}-d-1)/m]$. Further let $n = [d+1+(\bar{q}-d-1)/m]$. Using (3.4), we obtain

$$\begin{bmatrix} \gamma_u(0) \\ \gamma_u(1) \\ \vdots \\ \gamma_u(q^*) \end{bmatrix} = A_m^d(q) \begin{bmatrix} \gamma_w(0) \\ \gamma_w(1) \\ \vdots \\ \gamma_w(q) \end{bmatrix}.$$

Now increase the length of the L.H.S. vector to $n+1$ by appending $n-q^*$ zeros to it. Similarly extend the R.H.S. vector to length $\bar{q}+1$ by appending $\bar{q}-q$ zeros to it. Then we have

$$\begin{bmatrix} \gamma_u(0) \\ \gamma_u(1) \\ \vdots \\ \gamma_u(q^*) \\ 0 \\ \vdots \\ 0 \end{bmatrix} = A_m^d(\bar{q}) \begin{bmatrix} \gamma_w(0) \\ \gamma_w(1) \\ \vdots \\ \gamma_w(q) \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (3.5)$$

We call the LHS vector of (3.5) γ_u^* and the R.H.S. γ_w^* . Now since $\gamma_u(q^*)$ does not depend on any $\gamma_w(k)$ for kq , the matrix $A_m^d(\bar{q})$ can be partitioned as

$$A_m^d(\bar{q}) = \left[\begin{array}{c|c} A_m^d(q) & \\ \hline 0 & \beta \end{array} \right].$$

Consider the submatrix β which consists of the last $(\bar{q}-q)$ columns of $A_m^d(\bar{q})$. Because $\bar{q}-q > n$ by assumption, there exists a nonzero element, γ^n , of the null space of β . This implies that we can find an element of the null space of $A_m^d(\bar{q})$,

call it γ^k , which has a nonzero element beyond the $(q+1)$ st position. In particular, we let

$$\gamma^k = \begin{bmatrix} \mathbf{0}'_{q+1} \\ \gamma^n \end{bmatrix}$$

where we recall that $\mathbf{0}_{q+1}$ is a $1 \times (q+1)$ zero vector. Now the vector $A_m^d(\bar{q})(\gamma_w^* + \gamma^k)$ will equal γ_u^* . Since

Now the vector $A_m^d(\bar{q})(\gamma_w^* + \gamma^k)$ will equal γ_u^* . Since

$$\gamma_w^* + \gamma^k \quad (3.6)$$

has a nonzero element beyond the $q+1$ st position it is a candidate to define an IMA (d, q') model with $q' > q$. However, we need also show that there is a vector of form (3.6) which, upon dividing through by its first element, will fall in the admissible region for autocorrelations of an IMA $(d, \bar{q})$ model. By using lemma 3.1 we argue that we can. First we see that, after division, the autocorrelations calculated from γ_w^* are trivially in the admissible regions of an IMA (q') model since γ_w^* is admissible for an IMA (q) model. Second by shrinking γ^k , that is by multiplying it by a small nonzero scalar, we can make it close enough to, but unequal to $\mathbf{0}$ so that $\gamma_w^* + \gamma^k$ is close enough to γ_w^* so that it also is in the admissible region. In particular, using the notation of lemma 3.1, if the absolute value of each element of γ^k is less than $1/4\bar{q}$ then $\gamma_w^* + \gamma^k$ is guaranteed to be admissible. This is a contradiction. Hence q is not the upper bound. $\square$

4. CHANGE OF THE AUTOREGRESSIVE ORDER IN AR (p) AND ARIMA (p, d, q) MODELS AS A RESULT OF AGGREGATION

The AR (p) model for a series x_t provides a stochastic representation for x_t of form

$$\phi_p(B)x_t = a_t \quad (4.1)$$

Here $\phi_p(B)$ is equal to $(1 - \phi_1 B - \dots - \phi_p B^p)$, and is a polynomial in B with roots outside the unit circle. The $\{a_t\}$ is a series of iid random variables with mean 0 and finite variance. Of central importance to the study of these models is the fact that the autocovariances of x_t , $\gamma_x(k)$, for $k=0, 1, \dots$, follow the difference equation $\phi_p(B)\gamma_x(k) = 0$, for $k > 0$, where B operates on k .

In general, the temporal aggregates of a purely AR (p) process will have mixed autoregressive moving average representation. This has been pointed out by many authors including Amemiya and Wu (1972) and Wei (1978). Amemiya and Wu (1972) give the model for the m th order aggregate as an ARMA (p, q^*) process

$$\prod_{i=1}^p (1 - r_i^m B)y_t = \theta_{q^*}(B)c_t \quad (4.2)$$

for all t , where r_i^{-1} is the i th root of $\phi_p(B)$, $\theta_{q^*}(B)$ is a polynomial in B of order $q^* = [p+1 - (p+1)/m]$, with its roots outside the unit circle, and c_t is a white noise zero mean process.

The above authors fail to point out that the order given in (4.2) is only maximum. Whenever a series is to be modeled by a stochastic difference equation of form (4.2) it is necessary that $\prod_{i=1}^p (1 - r_i^m B)$ and $\theta_q(B)$ share no roots. If a root is shared by both polynomials then it may be cancelled from both sides and a simpler representation given. We are able to provide the general conditions when this occurs, and give, therefore, the true orders of the aggregate series representation. The result is given in the following theorem.

THEOREM 4.1. *Given $\phi_p(B)$ as in (4.1), let r_i^{-1} for $i = 1, \dots, p'$ be the distinct roots of $\phi_p(B)$, each with multiplicity s_i . Let b equal the number of distinct values r_i^m for $i = 1, 2, \dots, p'$. Further partition the numbers s_i , $i = 1, \dots, p'$ into b distinct sets A_i such that s_k and $s_j \in A_i$ iff $r_k^m = r_j^m$. Then the order of the AR portion of the model for y_t in (4.2) which remains after cancellation will equal $p^* = \sum_{i=1}^b \max A_i$, where $\max A_i =$ the largest element in A_i . The resulting order of the MA portion of the model will equal $N = q^* - (p - p^*)$ where $q^* = [p + 1 - (p + 1)/m]$.*

PROOF. The proof is obtained first for the case when all of the roots r_i^{-1} are distinct, implying that $p' = p$, with $s_i = 1$ and that $p^* = b$. Because $\phi_p(B)\gamma_x(k) = 0$, for $k > 0$, we can write $\gamma_x(k)$ as a sum of complex and real exponentials

$$\gamma_x(k) = \alpha_1 r_1^k + \dots + \alpha_p r_p^k, \quad (4.3)$$

where complex r_i 's produce damped sinusoidal waves. Here the α_i 's are suitably chosen constants, none of which equal zero. By (2.2) the result of aggregation is that

$$\gamma_y(k) = (1 + B + \dots + B^{m-1})^2 \gamma_x(mk + (m-1)). \quad (4.4)$$

Upon substitution of (4.3), we obtain for $k \geq 1$

$$\gamma_y(k) = (1 + B + \dots + B^{m-1})^2 (\alpha_1 r_1^{km+(m-1)} + \dots + \alpha_p r_p^{km+(m-1)}).$$

Allowing this polynomial to operate on the exponent of $r_i^{km+(m-1)}$ gives

$$\gamma_y(k) = \sum_{i=1}^p \alpha_i (1 + r_i^{-1} + \dots + r_i^{-(m-1)})^2 r_i^{(m-1)} r_i^{km}. \quad (4.5)$$

We note that each term in (4.5) has a factor which is constant w.r.t. k of form

$$\alpha_i r_i^{(m-1)} (1 + r_i^{-1} + \dots + r_i^{-(m-1)})^2 \quad (4.6)$$

which is guaranteed to be nonzero. It is nonzero because the α_i are taken to be nonzero and the roots r_i^{-1} are outside the unit circle whereas the polynomial $(1 + x + \dots + x^{m-1})$ has its roots on the unit circle. Consequently we see that the order of the difference equation which is satisfied by (4.5) equals the number of distinct values r_i^m . That is, if r_i^m is equal to r_j^m for $i \neq j$ then the i th and j th constant terms of form (4.6) can be combined and the order of the difference equation is reduced by one. If there are p^* distinct values of r_i^m in (4.5) then lemma 4.4 given in the appendix implies that the lowest order of the difference equation satisfied by (4.5) will be equal to p^* . Therefore the ARMA model for

y_t will have an AR order equal to p^* . This implies that we have seen a reduction of order in (4.2) from p to p^* . By lemma 4.1 given in the appendix the reduction of MA order in (4.2) equals the reduction of AR order. The resulting order for the MA portion of the model for y_t is then equal to $q^* - (p - p^*)$. This proves the Theorem when the roots of $\phi_p(B)$ are distinct.

If the polynomial $\phi_p(B)$ has multiple roots then (4.3) becomes a sum of the form

$$\gamma_x(k) = \sum_{j=1}^{p'} (\alpha_{j,0} + \alpha_{j,1}k + \dots + \alpha_{j,s_j-1}k^{s_j-1})r_j^k$$

and (4.5) becomes

$$\begin{aligned} \gamma_y(k) = & \sum_{j=1}^{p'} (\alpha_{j,0}(1+r_j^{-1} + \dots + r_j^{-(m-1)})^2 + \dots \\ & + \alpha_{j,s_j-1}(1+r_j^{-1} + \dots + r_j^{-(m-1)})^2 k^{s_j-1}) r_j^{(m-1)} r_j^{km}. \end{aligned}$$

Thus the order of the difference equation satisfied by $\gamma_y(k)$ equals $\sum_{i=1}^b \max A_i = p^*$ and by lemma 4.1 given in the appendix the AR and MA orders of the model for y_t will be equal to p^* and $q^* - [p - p^*]$, respectively. $\square$

EXAMPLE 4.1. For an example of the use of theorem 4.1, consider aggregation, when $m=3$, of the following model for x_t :

$$(1 - \phi_1 B - \phi_2 B^2)x_t = a_t,$$

with parameters $\phi_1 = -(0.5)^{1/3}$, $\phi_2 = -(0.5)^{2/3}$, and $\sigma_a^2 = 1$.

The polynomial $\phi_p(B)$ is of order 2 so that the result (4.2) implies $q^* = 2$ and hence an ARMA (2, 2) for y_t . However, the roots of $\phi_p(B)$ are $r_1^{-1} = \{2[\cos(120^\circ) + i \sin(120^\circ)]\}^{1/3}$ and $r_2^{-1} = \{2[\cos(120^\circ) - i \sin(120^\circ)]\}^{1/3}$, so $p' = 2$, $s_1 = s_2 = 1$; but $r_1^3 = r_2^3 = \frac{1}{2}$ implies $b = 1$ and $A_1 = \{1\}$. Thus, $p^* = 1$ and $q^* - (p - p^*) = 1$. So by Theorem 4.1 the model for y_t should be an ARMA (1, 1).

Theorem 4.1 shows that the AR order of a purely AR (p) series can change under temporal aggregation. When the basic series model is of the mixed form, ARIMA, this statement is doubly true. However, under certain conditions the AR order of a given model can remain the same after aggregation. We first give the following definition.

DEFINITION 4.1. Given an ARIMA (p, d, q) model in (1.1). Let r_i^{-1} 's be the roots of its AR polynomial. The model is said to be without hidden periodicity of order m if $r_i^m = r_j^m$ implies $r_i = r_j$.

THEOREM 4.2. For models of the form ARIMA (p, d, q) where there is no hidden periodicity of order m , the order p is unchanged by aggregation. Further the roots of the AR polynomial of the aggregate series model are the m th powers of the roots of the basic series AR polynomial.

PROOF. The proof is similar to that of theorem 4.1 with minor modifications required to account for the fact that the model is nonstationary. We give the proof first in the case when all the roots of the AR polynomial, $\phi_p(B)$ are distinct. Then, we have the autocovariances, $\gamma_w(k)$, following the difference equation: $\phi_p(B)\gamma_w(k) = 0$, for $k \geq q + 1$. For $k \geq l = \max(0, q - p)$, we can write

$$\gamma_w(k) = \alpha_1 r_1^{k-l} + \cdots + \alpha_p r_p^{k-l}, \quad (4.8)$$

where the α_i are nonzero constants. From (2.2) we have that

$$\gamma_u(k) = (1 + B + \cdots + B^{m-1})^{2(d+1)} \gamma_w(km + (d+1)(m-1)).$$

Considering k large enough, in particular, k such that $km \geq l + (m-1)(d+1)$, then $\gamma_u(k)$ depends only on $\gamma_w(i)$, where $i \geq l$, so that we can write $\gamma_u(k)$ as equal to:

$$\begin{aligned} \gamma_u(k) &= (1 + B + \cdots + B^{m-1})^{2(d+1)} \sum_{i=1}^p \alpha_i r_i^{km-l+(d+1)(m-1)} \\ &= \sum_{i=1}^p \alpha_i r_i^{-l+(d+1)(m-1)} (1 + r_i^{-1} + \cdots + r_i^{-(m-1)})^{2(d+1)} (r_i^m)^k. \end{aligned} \quad (4.9)$$

Now each term in this expression has a constant, with respect to k , of form

$$\alpha_i^* = \alpha_i (1 + r_i^{-1} + \cdots + r_i^{-(m-1)})^{2(d+1)} r_i^{-l+(d+1)(m-1)}.$$

Since the roots of the polynomial $(1 + x + \cdots + x^{m-1})^{2(d+1)}$ are all on the unit circle it follows that the constant α_i^* will be nonzero if and only if α_i in (4.8) is nonzero. Thus the true orders of the difference equations satisfied by (4.8) and (4.9) are the same (by lemma 4.4 given in the appendix) and so will be the true orders of their AR portions. Further, by lemma 4.3 given in the appendix, $\gamma_u(k)$ of (4.9) satisfies the difference equation $\prod_{i=1}^p (1 - r_i^m B) \gamma_u(k) = 0$, meaning that the roots of the aggregate series AR polynomial are equal to r_i^{-m} as claimed.

When the roots, r_i^{-1} are not all distinct the expression for $\gamma_u(k)$ becomes a more general sum of exponentials and polynomial terms in $(r_i^m)^k$ and k . The proof follows by using exactly the same arguments. $\square$

5. CONCLUDING REMARKS

The results obtained in this paper have an interesting implication in time series model building. The fact that a purely autoregressive model becomes a mixed autoregressive moving average upon aggregation may explain the ubiquitous nature of ARMA or ARIMA models in applied time series analysis. The possible reduction of the orders of AR and MA through aggregation implies that a simpler model may be due to higher order of data aggregation. Since aggregation often leads to information loss, a researcher may wish to study its impact on parameter estimation and forecasting. When only aggregates are available, a researcher may even wish to infer a model for the underlying process. The relationships between a given time series and its aggregates as shown in this paper are useful for these studies.

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APPENDIX: SUPPORTING LEMMAS

The proofs of theorem 4.1 and theorem 4.2 are based on several properties of difference equations. They are summarized in the following lemmas.

LEMMA 4.1. *Let b_t and z_t be sequences of continuous random variables satisfying the nonhomogeneous difference equation $\phi_p(B)z_t = \theta_q(B)b_t$, where the polynomials $\phi_p(B)$ and $\theta_q(B)$ are of finite orders p and q . Then there exists exactly one irreducible difference equation: $\phi_{p_0}^*(B)z_t = \theta_{q_0}^*(B)b_t$ where the polynomials $\phi_{p_0}^*(B)$ and $\theta_{q_0}^*(B)$ share no roots. Further $p - p_0 = q - q_0 \geq 0$.*

To further develop the properties of linear homogeneous difference equations needed here, we review the effect that differencing operators of the form $(1 - rB)^s$ have on sequences y_k . Throughout this discussion, B is assumed to act on k .

LEMMA 4.2. *Sequences of forms $r^k k^j$ satisfy difference equations $(1 - rB)^s y_k = 0$ for $j = 0, 1, 2, \dots, (s-1)$.*

PROOF. First consider the sequence r^k . This sequence will satisfy the difference equation $(1 - rB)y_k = 0$, because $(1 - rB)r^k = (r^k - r(r^{k-1})) = 0$.

Now consider the sequence $y_k = k^j$. Let us show that this y_k satisfies the equation

$$(1 - B)^s y_k = 0;$$

so long as j is a fixed nonnegative integer less than s .

The proof is by induction on s . Clearly the statement is true for $s = 1$. Now assume the statement holds for $s - 1$. Then we will have

$$\begin{aligned} (1 - B)^s k^j &= (1 - B)^{s-1} (k^j - (k-1)^j) \\ &= (1 - B)^{s-1} \left\{ - \sum_{l=0}^{j-1} \binom{j}{l} k^l (-1)^{j-l} \right\}. \end{aligned}$$

Now in each term remaining in the expansion k is raised to a power less than $s - 1$. Therefore, by the induction hypothesis, each such term is reduced to zero by the operator $(1 - B)^{s-1}$.

Now if y_k is any sequence on k we have $(1 - rB)r^k y_k = (r^k y_k - r(r^{k-1} y_{k-1})) = r^k (1 - B)y_k$. Repeated application of this result shows that

$$(1 - rB)^s r^k y_k = r^k (1 - B)^s y_k.$$

Therefore, if $y_k = k^j$ for $j = 0, 1, \dots, s-1$, we have the result that $(1 - rB)^s r^k k^j = r^k (1 - B)^s k^j = 0$. $\square$

Now, for the general homogeneous linear difference equation,

$$f(B)y_k = 0,$$

the solutions to the equation constitute a vector space of sequences over the field of scalars. That is, if x_k and y_k satisfy the difference equation then so will $\alpha_1 x_k + \alpha_2 y_k$, where the α 's are scalars. This, together with the fact that difference operators commute, leads to the following lemma.

LEMMA 4.3. *Any sequence of form*

$$\sum_{i=1}^p \sum_{j=0}^{s_i-1} \alpha_{ij} r_i^k k^j$$

will satisfy the difference equation $\prod_{i=1}^p (1 - r_i B)^{s_i} y_k = 0$. In fact, the sequence $r_i^k k^j$ for $i = 1, 2, \dots, p$ and $j = 0, 1, \dots, (s_i - 1)$ forms a basis for the space of solutions to the above difference equation.

The proof of $\{r_i^k k^j\}$ being a basis closely parallels analogous results concerning the functions $e^{\alpha x} x^j$ which arise in the consideration of solutions of linear homogeneous differential equations; see Ince (1956, p. 136). Based on the linear independence of these $r_i^k k^j$, we have

LEMMA 4.4. *The lowest order linear homogeneous difference equation satisfied by the sequence*

$$\xi_k = \sum_{i=1}^P \sum_{j=0}^{s_i-1} \alpha_{ij} r_i^k k^j,$$

where $\alpha_{is-1} \neq 0$ is

$$\prod_{i=1}^P (1 - r_i B)^{s_i} y_k = 0,$$

the order of this equation being $s = \sum_{i=1}^P s_i$.

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